MASTER'S THESIS
(PHYSICS)

Evolution of Open Quantum Systems under Continuous Measurement of their Environment

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- *Open quantum systems*
- *Weak measurements*
- *Stochastic master equation*
- *Quantum trajectories*
- *Photo/homodyne detection*

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« La gloire de Dieu, c'est de cacher les choses; La gloire des rois, c'est de sonder les choses. »
— **Proverbes 25:2**

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CONTEXT AND MOTIVATIONS

This document is the report of an internship carried out as part of a Master's degree in fundamental and numerical physics (CompuPhys).
My internship has been conducted within the team of Theorical Physics — *équipe de Physique Théorique* ΦTh — of the *UTINAM Institute* in the building of the *Observatory of Besançon*.
It aims to study from an academical perspective at first, the subject of continuous measurement considering open quantum systems. The main objective was to explore a subject almost-unknown in details to either my supervisor or myself. We wanted to understand the possibilities and how to handle this domain properly. This work could then lead to other searches linked to continuous measurements.

By familiarizing ourselves with the topic, we decided to adopt this perticular frame of report. The structure of this document is meant to help any unfamiliar reader (about continuous measurement), to discover and explore its fundamentals. We wanted to make it as illustrative and pedagogical as possible, thus, simple examples will be used to graphically depict mathematical and physical concepts.

Introduction

"Quantum mechanics is the description of the behavior of matter and light in all details and, in particular, of the happenings on an atomic scale" said *R.Feynman* in one of his lectures [1]. In this framework, position is no longer a single well-defined value prior to measurement; instead, it is described by a probability density all across space. Consequently, the notion of a trajectory as a continuous succession of points in spacetime is lost in quantum dynamics. We will see further below how to define these trajectories in the case of our study, but one question remains paramount: how can we even define a quantum system?

In the ideal case, systems are treated as *closed*, *i.e.*, isolated from any external environment and interacting with nothing at all. This modelling allows quantum mechanics to describe dynamics of systems with unitary and deterministic terms (the state at time t is determined solely by knowing the state at time t_0) using the Schrödinger equation. [2] However in the physical world, systems do "feel" their surroundings, *i.e.*, they interact with their environment. This interaction should be taken into account to obtain a more complete and physically relevant representation; in doing so, we describe the so-called *Open Quantum System* (OQS) theory [3].

Useful theories have been developed through the last decades, allowing quantum description to be more and more complete. One of them being the Quantum Measurement Theory (QMT) [4], where measurement is fully described. Measuring a system has a tremendous impact on the quantum world, we will take time to explain why further below. However, it is the very role of any experiment to interact with a system in order to measure it. The inquiry of measuring a system without impacting its quantum properties too much leads to the concept of weak measurements.

By studying QMT, one can thus notice that measuring a quantum system alters it. The measurement usually interacts so strongly with the system that it collapses into a state — an eigenstate of the observable — that lost interesting quantum properties, *e.g.*, losing entanglement[1]. Or, quantum systems can be in superposition of states, taking *Schrödinger's cat* thought experiment: the cat can be alive and dead at the same time before interaction. Measuring the state of the cat will determine its fate, alive or dead, it is then "stuck" and cannot be written as a superposition again. The idea of **weak measurements** is to **measure the environment** coupled with the system. Instead of altering the state, we obtain pieces of information on the system. The measurements can even be conducted continuously, maximizing the amount of data the observer can get. These are the global ideas behind **Quantum Continuous Measurements** (QCM) [5].

This work is organized as follows: first section covers an academical progression from quantum closed systems up to the evolution of open quantum systems under continuous photodetection/homodyne detection; following the qubit and harmonic oscillator examples. The second part is a study of a system of interest on which we apply quantum continuous measurement. (Si on a le temps.)

[1] For example, $|\psi\rangle_{AB} = \frac{|00\rangle+|11\rangle}{\sqrt{2}} \xrightarrow{P_0} |00\rangle_{AB} = |0\rangle_A \otimes |0\rangle_B$ a product state (thus non-entangled).

I From closed systems to continuous measurements — in a nutshell

The examples and illustrations you can find in the following are made on `Julia` using `QuantumToolbox.jl` especially to simulate the dynamics.

I.i) Closed quantum systems

To describe a system, one can refer to the postulates of quantum mechanics [2] summarized here:

1. The state of a system is represented by a wavefunction $|\psi\rangle \in \mathcal{H}$ living in a Hilbert space.
2. Physical quantities $\mathcal{A}$ are associated with operators $\hat{A}$, called observables. In the context of finite dimension space, they are simply Hermitian operators.
3. The only possible outcomes of a measurement of $\mathcal{A}$ are the eigenvalues a of $\hat{A}$, defined by $\hat{A}\,|a\rangle = a\,|a\rangle$.
4. When measuring $\mathcal{A}$ on a normalized system $|\psi\rangle$, we know the probability to get one of the possible eigenvalues: $\mathbb{P}(a_n)$ (which depends on the spectrum of the observable).
5. If the measurement of the physical quantity $\mathcal{A}$ on the system in the state ψ gives the result a_n, the state of the system immediately after the measurement becomes the eigenvector ϕ_n.
6. The time evolution of the wavefunction $|\psi(t)\rangle$ is governed by the Schrödinger equation:
$$i\hbar\frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = \hat{H}(t)\,|\psi(t)\rangle$$
where $\hat{H}$ is the hamiltonian of the system.

The kind of dynamics a system follows is then unitary and deterministic. We can always solve, at least analytically, for the solution of the Schrödinger equation.

I.ii) Open Quantum Systems (OQS)

I.iii) Quantum Measurement Theory (QMT)

I.iv) Stochastic Master Equations (SME)

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